

Linear Regression with Multiple variables

Multiple Features

Previously there was only one feature "x" and the model was:

$$f_{w,b}(x) = wx + b$$

x → y
Size in ft² → Price

But Now:

Feature

X ₁ Size in ft ²	X ₂ No. of bedrooms	X ₃ No. of floors	X ₄ Age of home in yrs	Price in \$1000s
2104	5	1	45	460
1416	3	2 (ii)	40	232
1534	3	2	30	315
852	2	1	36	178
...	...	...	...	...

- $X_j = j^{\text{th}}$ feature (here $j = 1 \dots 4$) as we have four features

- $n = \text{no. of features (4)}$

- $\vec{x}^{(i)}$ = features of i^{th} training example (vector) → row vector

arrow is optional

↳ e.g. $\vec{x}^{(2)}$ will be: $\vec{x}^{(2)} = [1416, 3, 2, 40]$

- $x_j^{(i)}$ = value of feature j in i^{th} training example

↳ $x_3^{(2)} = 2 \rightarrow (i)$

Now model will be:

$$f_{w,b}(x) = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b$$

↳ e.g.:

$$f_{w,b}(x) = 0.1x_1 + 4x_2 + 10x_3 + (-2)x_4 + 80$$

↓ ↓ ↓ ↓ ↓
size #bedroom #floor years base price

→ in this you can interpret b as \$@ (80.k\$) as base price for house w/o any features. (no bedroom, floor, age)

→ can think of 0.1 as maybe for every additional sq. foot the price will increase by 0.1 (100\$).

→ by additional room price increase by 4000\$.

→ for each floor, increase by 10k\$.

→ for each years age, price decrease by 2000\$.

for n features:

$$f_{\vec{w},b}(\vec{x}) = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

$$\vec{w} = [w_1, w_2, w_3, \dots, w_n] \rightarrow \text{vector}$$

$b = \text{a number}$

$$\vec{x} = [x_1, x_2, x_3, \dots, x_n] \text{ vector}$$

Simply can be rewritten as:

$$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b = w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n + b$$

↑
dot product

↳ multi-input feature

multiplying
corresponding no.
from vectors

so

"Multiple Linear"
Regression